C.W

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System of Linear equation:

Linear equation:

- Variable max power → 1

Ex: $x+3=7$

$x+y=9$

$x+y+z=100$

- একাধিক equations একসাথে নিলে system of linear equation (minimum 2টি)

(*) $x+y+z=2$
$2x+y+z=3$
$2x+5y+z=1$; Solve the following linear eqn:

Sol:

By using Argumented matrix,

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 2 & 1 & 1 & 3 \\ 2 & 5 & 1 & 1 \end{array}\right]$$

Target:
1. 3rd row তে all 0.
2. 2nd ও ৩ প্রথম 2টি 0.

$$= \left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 4 & 0 & -2 \\ 0 & -4 & 0 & 2 \end{array}\right] \begin{array}{l} R_2' = R_3 - R_2 \\ R_3' = R_2 - R_3 \end{array}$$

$$= \left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \quad \begin{aligned} R_3'' &= R_2' + R_3' \\ R_2'' &= R_2' + R_3' \end{aligned}$$

$\therefore x + y + z = 2$

Let, $x = 1$ and $y = 2$ are independent.

$\therefore z = 0 - 1$ (Ans)

HW:

(*) $5x + 2y = 3$

$2x + 3y + z = 1$; solve that.

$x + 4y + 3z = 7$